

Saketh Velidimalla

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AI/ML engineer and data scientist focused on production AI quality systems, LLM evaluation, AI observability, RAG regression testing, and release-quality gates. At Johnson & Johnson, owned evaluation, reporting, and cross-system measurement layers for JAIDA, a ServiceNow-based enterprise RAG system serving ~140,000 internal users. Specializes in turning ambiguous AI behavior into measurable failure modes, dashboards, and leadership-ready quality signals.

Core Competencies

- **AI Evaluation & LLMOps:** LLM-as-judge, RAG regression testing, hallucination detection, retrieval-quality analysis, Arize Phoenix, release-quality gates
- **AI Observability & Analytics:** no-context detection, routing-gap analysis, containment metrics, escalation modeling, repeat-user analysis, cross-system case stitching
- **Data Engineering & MLOps:** Python, SQL, PySpark/Spark, Airflow DAGs, AWS EMR, Databricks, DynamoDB, S3, Redshift, Docker
- **Applied NLP & Retrieval:** RAG pipelines, semantic similarity, topic modeling, embeddings, Word2Vec, Precision-at-K / Recall-at-K

Professional Experience

- **Johnson & Johnson** **New Brunswick, NJ**
Data Scientist *Feb 2025–Present*
 - Owned evaluation, analytics, reporting, and observability layers for JAIDA, a ServiceNow-based enterprise RAG question-answering system serving ~140,000 internal users across HR, procurement, finance, case management, and content governance
 - Built a production LLM evaluation and observability loop using rule-based classification, AWS Bedrock / Llama 3.2 judging, keyword fallback, and Arize Phoenix traces to detect retrieval faults, no-context sessions, routing gaps, hallucination patterns, and latency anomalies across thousands of daily queries
 - Built an LLM-driven content regression pipeline for JAIDA, comparing generated answers against ground truth and retrieved ServiceNow knowledge-base context; used LLM-as-judge evaluation, rule-based checks, and fallback classifiers to detect answer drift, retrieval misses, hallucination risk, and low-quality source articles before release
 - Architected cross-system analytics with ServiceNow, Genesys, and business operations stakeholders, connecting AI assistant sessions, support redirects, case creation, live chat/call data, and survey feedback; surfaced a 50–60% downstream case conversion rate used in leadership prioritization
 - Designed failure-triage logic that separated RAG quality issues into retrieval, generation, routing, content-quality, and telemetry categories, improving root-cause analysis for production quality reviews
 - Designed an AI measurement layer with multi-table semantic models, session/query-grain KPIs, containment metrics, repeat-user analysis, and weekly leadership reporting for executive quality reviews
- **Hewlett Packard Inc.** **Spring, TX**
Data Science Engineer *Jul 2024–Nov 2024*
 - Optimized PySpark pipelines on AWS EMR (~25% processing time reduction via query optimization and repartitioning); orchestrated Airflow DAGs across S3 ingestion, PySpark transformation, Redshift warehousing, and Power BI reporting
 - Translated raw hardware telemetry signals into engineering KPIs via advanced DAX; outputs used by hardware, software, and chip design teams to drive cross-functional design decisions
- **ECrent Worldwide Company** **Remote**
Machine Learning Engineer *Jul 2021–Jun 2022*
 - Built a real estate recommendation system that matched users to relevant property listings based on listing text and structured attributes; fine-tuned Word2Vec embeddings, ranked candidates by cosine similarity, served recommendations through a Flask API, and evaluated relevance with Precision-at-K and Recall-at-K

Independent Projects

- **roastrank-v2:** LLM evaluation and benchmarking platform with rubric-based scoring, regression testing, six-stage output validation, failure labeling, and audit logging. FastAPI, Docker, Hugging Face.
- **Buddy:** Agentic AI runtime prototype with layered memory, retrieval routing, context assembly, local LLM fallback, and unit-tested prompt/context behavior. Python, Anthropic API.
- **AI Digest:** Automated AI news and production-research digest system posting curated summaries to Slack on a GitHub Actions schedule. Gemini, Slack Block Kit, Python.

Education & Certifications

- **Arizona State University** **Phoenix, AZ**
MS Information Technology (Data Science) *Aug 2022–May 2024*
- **Completed:** Codex in Action, Building with Codex API, Agentic AI specialization (Andrew Ng); LLM Engineering: Foundations & Production (Ed Donner)
- **In Progress:** AWS Certified ML Specialty (Q2 2026); Databricks Generative AI Engineer Associate (Q3 2026)